



CENTRE FOR ADVANCED STUDIES
Dr. APJ Abdul Kalam Technical University, Lucknow

M.TECH - MECHATRONICS

Effective from 2018-19
Program Structure

The Centre for Advanced Studies (Dr. A.P.J. Abdul Kalam Technical University, Lucknow, U.P.) is starting Master of Technology in Mechatronics from the academic session 2018-19. The curriculum has been developed considering the present and future needs of industry and higher education. The Centre will facilitate both industry ready and research based ambience to students with world class e-library, renowned faculty members to achieve academic excellence and other services.

This is a program aimed to provide the student with in-depth knowledge of domains in the field of Mechatronics.

M. Tech in Mechatronics. It is a multi-disciplinary study dealing with the integration of mechanical devices, actuators, sensors, electronics, intelligent controllers and computers. It is unique balance of rigorous fundamentals and engaging real-world applications in Mechatronics Engineering which provide the students the opportunity to design, build, operate and maintain machinery with mechanical, electronic/electrical and computer control systems.

The course structure will help students to develop knowledge and skill in the following proportionate:

90% Technical /Research	Industrial automation and robotics, Sensors and actuators, Control systems, Machine vision, Microcontrollers and embedded systems. Advanced microcontrollers, Mechatronics systems design, Machine tool control and condition monitoring, Flexible manufacturing system, Smart materials for mechatronics, Additive manufacturing and tooling, Vetronics, Finite element method, Bio mechatronics, and Intelligent visual surveillance systems, IOT and smart manufacturing.
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	Excellent Visionary Skill that focus on scalability, cost effectiveness and implementation ease.
10% Business Process & Managerial Practices	Know the Business Dynamics, Business Processes and good planning, ability to work with all management level and resolve issues, Business need with automation Requirement. Consulting Skill, Communication Skill, Legal Understanding.

The students of Mechatronics program will attain following objectives:

- a) Students will understand the evolving Mechatronics systems from their underlying physical principles and properties.
- b) Students will design Mechatronics systems by applying underlying mathematical principles, supporting software for engineering model preparation and analysis.
- c) Students will be proficient in the systematic explorations of alternatives for Mechatronics systems design.
- d) Students will be proficient in the use of communications (oral presentations and written reports) to articulate their ideas effectively.
- e) Students will be prepared for the continuing learning and self-improvement necessary for a productive career in Mechatronics.

Program Structure:

M.Tech course is a full time two year program and classes will be held on all working days. The Program Structure has been designed such that the students shall study core subjects of Mechatronics Engineering.

In Semester 1, students shall study fundamental core courses of Mechatronics Engineering while in the next two semesters students shall study in depth Advanced core and elective subjects. Semester IV is Thesis/ Dissertation.

Every core course consists of lecture (L) hours, tutorial (T) hours and practical (P) hours. Elective courses consist of Lecture (L) hours only. The credit I for a course is dependent on the number of hours of instruction per week in that course, as given below:

- (1) 1h/week of Lecture (L) = 1 credit
- (2) 2h/week of Practical (P) = 1 credit
- (3) 1h/week of Tutorial(T) = no credit
- (4) Credit I for a Theory course = No. of hours of lectures per week = L
- (5) Credit I for a Lab course = $\frac{1}{2}$ * No. of hours of lab per week = P

Course Credit Distribution

Category	Core Courses	Elective Courses	Labs	Project	Total Credit
Credits	21	15	7	17	60

SEMESTER	COURSE TYPES	CREDITS	
Semester I			
1	Core Course	(3 credit * 5 course) = 12	TOTAL COURSE = 5 TOTAL SEM CREDIT = 18
	Elective Course	(3 credit * 1 course) = 3	
2	Core Course Lab and seminar	1 credit * 3 course = 3	
Semester II			
2	Core Courses	3 credit * 2 course = 6	TOTAL COURSE = 5 TOTAL SEM CREDIT = 17
2	Elective Course	3 credit * 3 course = 9	
	Core Course Lab and comprehensive viva-voce	1 credit * 2 course = 2	
Semester III			
3	Core Courses	3 credit * 1 course = 3	TOTAL COURSE = 3 TOTAL SEM CREDIT = 12
3	Industrial Training/Summer Research intern	2 credit * 1 course = 2	
3	Thesis/Dissertation-I/Seminar	7 credit	
Semester IV			
4	Thesis/ Dissertation-II	13 credit	TOTAL COURSE = 1 MAJOR PROJECT TOTAL SEM CREDIT = 13
TOTAL PROGRAM CREDIT			60

SEMESTER –I

S No.	Course Code	Subject	Periods (Hrs/week)			Credits
1	MMTR-101	Advanced Engineering Mathematics	3	0	0	3
2	MMTR-102	Sensors and Actuators	3	0	0	3
3	MMTR-103	Control System: Modeling, Design and Analysis	3	0	0	3
4	MMTR-104	Microcontrollers and Embedded Systems	3	0	0	3
5	MMTR-105/ MMTR-106	Concepts of Mechanical Engineering/ Concepts of Electronics and Electrical Engineering	3	0	0	3
6	MMTR-151	Lab-I Sensors, Drives and Control lab	0	0	2	1
7	MMTR-152	Lab-II Microcontroller and Embedded systems lab	0	0	2	1
8	MMTR-153	Seminar	0	0	2	1
		TOTAL	15	0	6	18

SEMESTER –II

S No.	Course Code	Subject	Periods (Hrs/week)			Credits
1	MMTR-201	Industrial Automation	3	0	0	3
2	MMTR-202	Industrial Robotics	3	0	0	3
3		Elective-I	3	0	0	3
4		Elective- II	3	0	0	3
5		Elective- III	3	0	0	3
6	MMTR-251	Lab-III (Robotics and Automation lab)	0	0	3	1

7	MMTR-252	Seminar/ Comprehensive Viva-voce (At end of II Semester)	0	0	3	1
		TOTAL	15	0	6	17

SEMESTER –III

S No.	Course Code	Subject	Periods (Hrs/week)			Credits
1	MMTR-301	Research Methodology and Technical Writing	3	0	0	3
2	MMTR-351	Research/Industrial Training (Attended after II semester- Approx. 4-6 weeks)	0	0	4	2
4	MMTR-352	Thesis/Dissertation- I	0	0	8	7
5		Teaching/Lab Assignments & evaluation	0	0	8	0
		TOTAL	6	0	12	12

SEMESTER –IV

S No.	Course Code	Subject	Periods (Hrs/week)			Credits
1	MMTR-451	Thesis / Dissertation - II	0	0	30	13
		TOTAL	0	0	30	13

Elective Subjects-

Elective- (I-III)

S. No.	Course code	Subject
1	MMTR-211	Mechatronics Systems Design
2	MMTR-212	PC based automation

3	MMTR-213	Signal Processing in Mechatronics Systems
4	MMTR-214	Control of Mechatronic Systems
5	MMTR-215	Machine tool control and condition monitoring
6	MMTR-216	Mechatronics in Metrology
7	MMTR-217	Machine vision
8	MMTR-218	Design of Mechanisms and machines
9	MMTR-219	Artificial Intelligence & Expert System in Automation
10	MMTR-220	Modeling and Simulation of Mechatronic Systems
11	MMTR-221	Advanced microcontroller and embedded systems
12	MMTR-222	Finite element method
13	MMTR-223	Mechatronics in CNC
14	MMTR-224	Theory of Machining
15	MMTR-225	Computer Aided Manufacturing
16	MMTR-226	Flexible Manufacturing System
17	MMTR-227	Rapid Prototyping
18	MMTR-228	MEMS and NEMS
19	MMTR-229	Smart materials for mechatronics applications
20	MMTR-230	Optimization Techniques & Design of Experiments
21	MMTR-231	Additive Manufacturing & Tooling
22	MMTR-232	Vetronics
23	MMTR-233	Computational fluid dynamics

24	MMTR-234	Intelligent visual surveillance systems
25	MMTR-235	Bio-Mechatronics
26	MMTR-236	Internet of Things and Smart Manufacturing

Labs

S No.	Lab code	Lab Name
1	MMTR1-151	Sensors, Drives and Control lab
2	MMTR1-152	Microcontroller and Embedded systems lab
3	MMTR1-153	Seminar
4	MMTR1-251	Robotics and Automation lab
5	MMTR1-252	Seminar/ Comprehensive Viva-voce (At end of II Semester)
6	MMTR-351	Research/Industrial Training (Attended after II semester- Approx. 4-6 weeks)
7	MMTR-352	Thesis/Dissertation- I
8	MMTR-451	Thesis / Dissertation - II



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SYLLABUS FOR M.TECH (MECHATRONICS) Effective from 2018-19

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1	Semester- I	11
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MARKS DISTRIBUTION

Following is marks distribution for courses depending upon the credit of the courses:

(1 credit = 50 marks)

Credit	Type of course	Max. Marks	Major (End semester examination)	Performance evaluation		Lab evaluation
				Minor I & II	Internal Assessments (Attendance, Quiz, Seminar & Assignments)	
1	Lab course	50	25	25		-
2	Theory course	100	50	30 (15 each)	20	-
3	Theory course	150	75	40 (20 each)	35	-
4	Theory course with lab	200	75	40 (20 each)	35	50
4	Thesis / dissertation-I	350	200	150		-
5	Thesis / dissertation-II	650	400	250		-

Note: Maximum Marks is fixed for all the courses. Only internal assessments and lab evaluation marks distribution may vary as per the instructions communicated by the assigned Faculty. Refer to M.Tech Ordinance & Regulation for Assessment Procedures and Performance Evaluation.

SEMESTER-I
CORE COURSES

Course Code: MMTR-101,
Course Name: Advanced Engineering Mathematics,
Credit: 3

Syllabus:

Prerequisite NIL

Linear Algebra: Matrix algebra; basis, dimension and fundamental subspaces; solvability of $Ax = b$ by direct Methods; orthogonality and QR transformation; eigenvalues and eigenvectors, similarity transformation, singular value decomposition, Fourier series, Fourier Transformation, FFT.

Vector Algebra & Calculus: Basic vector algebra; curves; grad, div, curl; line, surface and volume integral, Green's theorem, Stokes's theorem, Gauss-divergence theorem.

Differential Equations: ODE: homogeneous and non-homogeneous equations, Wronskian, Laplace transform, series solutions, Frobenius method, Sturm-Liouville problems, Bessel and Legendre equations, integral transformations; PDE: separation of variables and solution by Fourier Series and Transformations, PDE with variable coefficient.

Numerical Technique: Numerical integration and differentiation; Methods for solution of Initial Value

Problems, finite difference methods for ODE and PDE; iterative methods: Jacobi, Gauss-Siedel, and successive over-relaxation.

Complex Number Theory: Analytic function; Cauchy's integral theorem; residue integral method, conformal mapping. Statistical Methods: Descriptive statistics and data analysis, correlation and regression, probability distribution, analysis of variance, testing of hypothesis.

Text Books:

1. H. Kreyszig, "Advanced Engineering Mathematics", Wiley, (2006).
2. Gilbert Strang, "Linear Algebra and Its Applications", 4th edition, Thomson Brooks/Cole, India (2006).
3. J. W. Brown and R. V. Churchill, "Complex Variables and Applications", McGraw-Hill Companies, Inc., New York (2004).
4. J. W. Brown and R. V. Churchill, "Fourier Series and Boundary Value Problems", McGraw-Hill Companies, Inc., New York (2009).
5. G. F. Simmons, "Differential Equations with Applications and Historical Notes", Tata McGraw-Hill Edition, India (2003).
6. S. L. Ross, "Differential Equations" 3rd edition, John Wiley & Sons, Inc., India (2004).
7. K. S. Rao, "Introduction to Partial Differential Equations", PHI Learning Pvt. Ltd (2005).
8. R. Courant and F. John, "Introduction to Calculus and Analysis, Volume I and II", Springer-Verlag, New York, Inc. (1989).
9. K. Atkinson and W. Han, "Elementary Numerical Analysis" 3rd edition, John Wiley & Sons, Inc., India

(2004).

10. R. A. Johnson and G. K. Bhattacharya, “Statistics, Principles and Methods”, Wiley (2008).

Course Code: MMTR-102,

Course Name: Sensors and Actuators,

Credit: 3

Syllabus:

Prerequisite: NIL

Brief overview of measurement systems, classification, characteristics and calibration of different sensors.

Measurement of displacement, position, motion, force, torque, strain gauge, pressure flow, temperature sensor sensors, smart sensor. Optical encoder, tactile and proximity, ultrasonic transducers, opto-electrical sensor, gyroscope. Principles and structures of modern micro sensors, micro-fabrication technologies: bulk micromachining, surface micromachining, LIGA, assembly and packaging.

Pneumatic and hydraulic systems: actuators, definition, example, types, selection. Pneumatic actuator. Electro-pneumatic actuator. Hydraulic actuator, control valves, valve sizing valve selection. Electrical actuating systems: solid-state switches, solenoids, voice coil; electric motors; DC motors, AC motors, single phase motor; 3-phase motor; induction motor; synchronous motor; stepper motors. Piezoelectric actuator: characterization, operation, and fabrication; shape memory alloys.

Text Books

1. John G. Webster, Editor-in-chief, “Measurement, Instrumentation, and Sensors Handbook”, CRC Press (1999).

2. Jacob Fraden, “Handbook of modern Sensors”, AIP Press, Woodbury (1997).

3. Nadim Maluf, “An Introduction to Microelectromechanical Systems Engineering”, Artech House Publishers, Boston (2000).

4. Marc Madou, “Fundamentals of Microfabrication”, CRC Press, Boca Raton (1997).

5. Gregory Kovacs, “Micromachined Transducers Sourcebook”, McGraw-Hill, New York (1998).

6. E. O. Deobelin and D. Manik, “Measurement Systems – Application and Design”, Tata McGraw-Hill (2004).

7. D. Patranabis, “Principles of Industrial Instrumentation”, Tata McGraw-Hill, eleventh reprint (2004).

8. B. G. Liptak, “Instrument Engineers’ Handbook: Process Measurement and Analysis”, CRC (2003).

Course Code: MMTR-103,

Course Name: Control System: Modeling, design and Analysis,

Credit: 3

Prerequisite NIL**Syllabus:**

Introduction and need for Control Systems, Block Diagram reduction, Signal flow graph, Transfer Function Model, Mathematical Model and State Space Model.

Time response of First & Second order systems – Time domain specifications - steady state errors and error constants – Routh Hurwitz criterion – Root locus – Root locus approach to control system design – Lead, Lag, Lag-Lead Compensation using time domain analysis.

Feedback systems – Block Diagram – Definition of Process variable, Set-point, Manipulated variable and Final control elements, Characteristics of on-off, P, PI, PD and PID Controllers – Implementation issues of

PID Controller – Modified PID Controller – Tuning of controllers.

Bode Plot – Polar Plot – Nyquist stability criterion – Stability analysis – Experimental determination of Transfer Functions – Control system design using Frequency domain analysis - Lead, Lag, Lag-Lead Compensation using frequency domain analysis.

Servo motor – Mathematical Modelling of Servo Motor – Analysis of Servo motor system using Routh

Hurwitz criterion, Root locus, Bode Plot, Polar Plot and stability analysis – Implementation of P, PI, PD and PID controllers for servo motor and analysis.

References:

1. K. Ogata, Modern controls engineering “Prentice Hall of India Pvt. Ltd., New Delhi, 2005.
2. B.C. Kuo, “Automatic Control Systems”, Prentice Hall of India Pvt. Ltd., New Delhi, 2004
3. I.J. Nagrath and Gopal. “Control system engineering”, new age international (P) Ltd., 2006.
4. M. Nakamura. S. Gata & N. Kyura, Mechatronic servo system control. Springer 2009.

Course Code: MMTR-104,

Course Name: Microcontrollers and Embedded Systems,

Credit: 3

Syllabus:**Prerequisite NIL**

Introduction to Embedded Systems and microcomputers: Introduction to Embedded Systems, Embedded System Applications, Block diagram of embedded systems, Challenges for Embedded system Design, Evolution of computing systems and applications. Classification of the processors used in embedded design, Basic Computer architecture: Von- Neumann and Harvard Architecture. Basics Processing Cores – RISC & CISC. Computing performance, Throughput and Latency, Microcomputer applications to embedded systems and Mechatronics.

Microcontroller:

Introduction to Microcontroller and its families, Criteria for Choosing Microcontroller. Microcontroller Architecture, Programming model, Addressing modes, Instruction sets, Assembly and C programming for Microcontroller, I/O programming using assembly and C language, Interrupt Controller, I/O interfacing, Timers, Serial Communication, LCD Controller. Different types of Sensors.

Microcontroller Interfacing: Introduction to Microcontroller Interfacing and applications: case studies: Display Devices, Data Acquisition Systems, controllers and Drivers for DC, Servo and Stepper Motor.

Introduction to MSP 430: Memory Model, Overview of Instructions, I/O Ports, Timers and Interrupts.

Introduction to Advanced Embedded Processor and Software: ARM Processor, Real Time Operating System (RTOS), Embedded C.

References:

1. Introduction to Embedded Systems: Shibu K V, McGRAW Hill Publications.
2. Embedded Systems: Raj Kamal, TATA McGRAW Hill Publications
3. Computer System Architecture: M. Morris Mano.
4. PIC Microcontrollers and Embedded Systems: M. A. Mazidi, R.D. Mckinlay and D. Casey, Pearson Publications
5. Programming and Customizing the PIC Microcontroller: M. Predko, McGRAW Hill Publications.
6. Embedded C Programming and Microchip PIC: R. Barnett, L. O'Cull and S. Cox.

Course Code: MMTR-105,

Course Name: Concepts of Mechanical Engineering,

Credit: 3

Syllabus:

Prerequisite- NIL

Mechanisms: Definition – Machine and Structure – Kinematic link, pair and chain – classification of Kinematic pairs – Constraint & motion - Degrees of freedom, Slider crank mechanisms – Inversions – applications.

Power Screws: Forms of threads, multiple threads, Efficiency of square threads, Trapezoidal threads, Stresses in screws, Design of screw jack.

Friction drives: Introduction, belt and rope drives, open and crossed belt drives, power transmission.

Gearing and cams: Gear profile and geometry-nomenclature of spur and helical gears, law of gearing simple and compound gear trains. Cam profile-different types of followers.

Fluid mechanics: Fluid and continuum, Physical properties of fluids, Rheology of fluids, Pressure-density-height relationship, manometers. Types of fluid flows, Bernoulli's equation and its applications, Equation of motion for laminar flow through pipes.

Strength of materials: Introduction, normal stress and strain, shear stress and strain, stresses on inclined sections, strain energy, impact loads and stresses, state of plane stress, principal stress and strain, maximum shear stress, three-dimensional state of stress & strain, equilibrium equations, theories of failure.

Thermodynamics: Fundamental laws, Temperature, Enthalpy, Entropy. Internal combustion engines.

References:

1. Joseph Edward Shigley, Charles R. Mischke, "Mechanical Engineering Design" McGraw Hill International Edition, 2008.
2. Juvinall, R.C., Fundamentals of Machine Component Design, John Wiley and Sons, 1994.
3. Spottes, M.F., Design of Machine elements, Prentice-Hall India, 1994.

4. Deutschman, D., Michels, E., Machine Design Theory and Practice, Macmillan, 1992
5. Collens, J.A., “Failure of Materials in Mechanical Design”, John Wiley & Sons, 1991.
6. Fuchs H.O., “Metal fatigue in Engineering”, John Wiley & Sons, 1991.
7. Thermodynamics, P.K. Nag Mcgraw Hill Education.

Course Code: MMTR-106,

Course Name: Concepts of Electronics and Electrical Engineering,

Credit: 3

Prerequisite- NIL

Syllabus:

Digital- number system, logic gates and their applications, K-map, Combinational Logic circuits, Synchronous and Asynchronous Sequential logic circuits, Registers and counters, Memory and programmable logic.

Oscillators, Brief introduction of Transformer, Solid state devices- SCR, IGBT, Converters & Invertors, Variable speed drives (AC & DC). Test and measuring instruments-Measurement of voltage, current, frequency and power using multi meters, oscilloscopes, recorders, data loggers, signal sources, analysers and printers.

Microprocessor (8085/8086) architecture and its operations, pins description, programming model, basic interfacing concepts. Memory Interfacing, Programmable Peripheral Interfacing, Programmable I/O, Programmable Controllers, Programmable Timers, Data acquisition Interfacing: ADC, DAC, Serial and parallel data Communication interfacing.

References:

1. Millman Halkias, Integrated Electronics, Tata McGraw-Hill
2. M. Morris Mano, Digital Design, McGraw-Hill
3. Gayakwad, Operational Amplifier-Linear Integrated Circuits, Prentice Hall India
4. Mohammad H. Rasid, Power Electronics, Prentice Hall India
5. Helfrick A.D and Cooper .W. D., Modern Electronic Instrumentation and Measurements Techniques, Prentice Hall 2008.
6. Leach, Digital Principles and Applications, ed. 7, TMH 2008.
7. Mandal, Digital Electronics: Principles and Applications, TMH 2009.
8. 8086 Microprocessors and Interfacings: D. Hall, TATA McGRAW Hill.
9. The Intel Microprocessors: B. Brey, Prentice Hall Publications.

SEMESTER- II CORE COURSES

Course Code: MMTR-201,
Course Name: Industrial Automation,
Credit: 3

Syllabus:

Prerequisite- NIL

Introduction: Automation in Production System, Principles and Strategies of Automation, Basic Elements of an Automated System, Advanced Automation Functions, Levels of Automations. Production Economics.

Detroit-Type Automation: Automated Flow lines, Methods of Work part Transport, Transfer Mechanism, and Buffer Storage.

Material handling and Identification Technologies: The material handling function, Types of Material Handling Equipment, Analysis for Material Handling Systems, Design of the System, Conveyor Systems, Automated Guided Vehicle Systems. Automated Storage Systems: Storage System Performance, Automated Storage/Retrieval Systems, Work-in-process Storage, Product identification system: Barcode, RFID etc.

Automated Assembly Systems: Design for Automated Assembly, Types of Automated Assembly Systems. Different control technologies in automation:

Automated Inspection and Testing: Inspection and testing, Automated Inspection Principles and Methods, Sensor Technologies for Automated Inspection, Coordinate Measuring Machines, Other Contact Inspection Methods, Machine Vision, Other optical Inspection Methods.

References:

1. “Automation, Production Systems and Computer Integrated Manufacturing”- M. P. Grover, Pearson Education.
2. “Computer Based Industrial Control” – Krishna Kant, EEE-PHI
3. Principles and Applications of PLC – Webb John, Mcmillan 1992
4. “An Introduction to Automated Process Planning Systems” – Tiess Chiu Chang & Richard A. Wysk
5. “Anatomy of Automation” – Amber G.H & P.S. Amber, Prentice Hall.

Course Code: MMTR-202,
Course Name: Industrial Robotics,
Credit: 3

Syllabus:

Prerequisite- NIL

Fundamentals of robotics. Introduction, construction and applications.

The robot and its peripherals: Control systems and components, robot motion analysis and control, end effectors, feedback systems, encoders kinematics, homogeneous coordinates solution of the inverse kinematic problem, multiple solutions, jacobian, work envelopes. Trajectory planning, Joint Interpolated Trajectory, Link joints and their Manipulator dynamics and force control.

Sensors in Robotics: Developments in sensor technology, sensory control Vision, ranging, laser, acoustic, tactile.

Programming Language: Industrial robot programming languages. Mobile robots, robot avoiding system, walking devices. Robot programming environment.

Robot applications: Application of robots in surgery, Manufacturing industries, space and underwater.

Humanoid robots, Micro robots, Social issues and Future of robotics.

References:

1. K.S Fu, R.C. Gonzalez, C.S.G. Lee, Robotics, McGraw Hill, 1987.
2. Y. Koren, Robotics for Engineers, McGraw Hill, 1985.
3. J.J. Craig, Robotics, Addison-Wesley, 1986.
4. Saeed B. Niku, "Introduction to Robotics – Analysis, Systems and Application": PHI 2006.
5. Richard D, Klafter, Thomason A Chmiel Owski, Michel Nagin "Robotics Engg-an Integrated Approach" PHI 2005.
6. R.K. Mittal & I.J. Nagrath, "Robotics & Control" TMH-2007.
7. Groover. M.P. Industrial Robotics, technology, programming and application Mc-Graw Hill 2012.
8. S. K. Saha, "Introduction to Robotics", Tata McGraw-Hill Publishing Company Ltd. (2008).

ELECTIVE- (I-I11)

Course Code: MMTR-211,

Course Name: Mechatronics Systems Design,

Credit: 3

Prerequisite- NIL

Syllabus:

Mechatronic systems, Key elements, Mechatronic design process, Application types, Interfacing issues, Man Machine Interfaces, Safety features, optimization of Mechatronic design, Fault diagnosis.

Mathematical models, Block diagram modelling, Analogy approach, Impedance diagrams, Models for Electrical, Mechanical, Electro-mechanical and Fluid systems, System Identification, Least square method, Closed loop identification, joint input/output identification, State estimators, Model Validation.

Simulation basics, Probability concepts in simulation, Discrete event simulation, Simulation Methodology, Queuing system model components, Continuous system modelling, Monte Carlo simulation, Analysis of simulation results, Simulation life cycle.

Mass-Spring-Oscillation and Damping system, Position Control of Permanent magnet DC motor using Hall sensor and optical encoder, Auto-control system for Green House Temperature, Transducer Calibration system, Strain Gauge Weighing system, Solenoid Force-Displacement Calibration system.

Automatic Washing Machine, Hard Drive control, Auto-focusing in Digital Cameras, Active suspension in vehicles, Visual Servoing models, Thermal cycle fatigue of a Ceramic plate, pH

Control system, De-icing temperature control system, Skip control of a CD player, Simulation of Rocket thrust control, Time delay Blower.

References

1. Devadas Shetty, Richard A.Kolkm, "Mechatronics system design, PWS publishing company, 2009.
2. Bolton, "Mechatronics – Electronic control systems in mechanical and electrical engineering, 2nd edition, Addison Wesley Longman Ltd., 2009.
3. Brian morriss, "Automated manufacturing Systems – Actuators Controls, sensors and Robotics", McGraw Hill International Edition, 2000.
4. Bradley, D. Dawson, N.C.Burd and A.J. Loader, "Mechatronics: Electronics in product and process", Chapman and Hall, London, 1999

Course Code: MMTR-212,

Course Name: PC based automation,

Credit: 3

Prerequisite- NIL

Syllabus:

Basic Computer Architecture, Components of a PC, Serial and Parallel Communications, Parallel port, OSI Model, RS232, USB, ISA, PCI, PXI, PCI Express, GPIB Protocols. LAN, WAN and MAN Networks, RS485, RS 422, LXI Protocols, Modbus, Field bus, Ethernet, CAN bus, SCADA and DCS.

Continuous and Discrete signals, Sampling theorem, Quantization, Sampling and Hold, ADC,DAC, Resolution and Sampling Frequency, Multiplexing of input signals, Single ended and differential inputs, Sampling of Multi-channel analog signals, Concept of Universal DAQ card, Timer, Counter and analog output in Universal DAQ card.

Algorithm, Flowchart, Variables & Constants, Expressions, Data types, Input output operations, Conditional Statements, Looping, Sub-programs/Functions, Arrays, Structures and Classes, Inheritance, Polymorphism, Debugging.

GUI, Graphical Programming, Data Flow techniques, Processing Data in GP, Loops and Structures, Event based & Schedule based operations, Global and Local Variables, File I/O operations, parallel processing of data, Virtual Instrument and control, VISA & SCPI.

References:

1. Morris Mano M., Computer System Architecture, Prentice Hall of India, Third Edition, 2002.
2. John P. Hayes, Computer Architecture and Organization, McGraw Hill International, Third Edition, 1998.
3. William Stallings, "Computer Organization and Architecture", VI Edition, Prentice Hall of India, 2003.
4. Krishna Kant, 'Computer based Industrial Control', Prentice Hall of India, 1997.
5. Gary Johnson, 'LabVIEW Graphical Programming', II Ed., McGraw Hill, 1997.
6. Sanjeev Gupta, 'Virtual Instrumentation using Labview' Tata McGraw Hill, 2004.
7. Jovitha Jeome 'Virtual Instrumentation using Lab View' PHI Learning Pvt Ltd, 2009.

Course Code: MMTR-213,

Course Name: Signal Processing in Mechatronics Systems,

Credit: 3

Prerequisite- NIL

Syllabus:

Discrete- Time Signals: Sequences; representation of signals on orthogonal basis; Sampling and Reconstruction of signals

Discrete systems: Z-Transform, Analysis of LSI systems, Frequency Analysis, Inverse Systems, Discrete Fourier Transform (DFT), Fast Fourier Transform algorithm, Implementation of Discrete Time Systems.

Analog filters: Ideal filter characteristics, low pass, high pass, band pass and band stop filters, notch filters, comb filters, all-pass filters, and inverse systems. Digital filters: Design of FIR and IIR filters.

Introduction to DSP Processors. Introduction to various Texas processors such as TMS320C6713, TMS320C6416, DM6437 Digital Video Development Platform with Camera, DevKit8000 OMAP3530 Evaluation Kit.

Introduction to multi-rate signal processing. Application of DSP to Speech and Radar signal processing.

References:

1. S. K. Mitra, Digital Signal Processing: A computer-Based Approach, 3/e, TMcHl, 2006.
2. A. V. Oppenheim and R. W. Shafer, Discrete-Time Signal Processing, Prentice Hall India, 2/e, 2004.
3. J. G. Proakis and D. G. Manolakis, Digital Signal Processing: Principles, Algorithms and Applications, 4/e, Pearson Education, 2007.
5. V.K. Ingle and J.G. Proakis, "Digital signal processing with MATLAB", Cengage, 2008.
6. T. Bose, Digital Signal and Image Processing, John Wiley and Sons, Inc., Singapore, 04.
7. L. R. Rabiner and B. Gold, Theory and Application of Digital Signal Processing, PH, 2005.
8. A. Antoniou, Digital Filters: Analysis, Design and Applications, Tata McH, 2003.

Course Code: MMTR-214,

Course Name: Control of Mechatronic Systems,

Credit: 3

Syllabus:

Prerequisite NIL

Modern control: Concept of states, state space model, different form, controllability, observability; pole placement by state feedback, observer design, Lunenburg observer, reduced order observer, observer based control.

Optimal control design: Solution-time criterion, control-area criterion, performance indices; zero steady state step error systems; modern control performance index: quadratic performance index, Ricatti equation.

Digital control: Sampling process, sample and hold, analog to digital converter, use of z-transform for closed loop transient response, stability analysis using bilinear transform and Jury method, digital control design using state feedback

Non-Linear Control System: Common physical non-linear system, phase plane method, system analysis by phase plane method, stability of non-linear system, stability analysis by describing function method, Liapunov's stability criterion, Popov's stability criterion.

Text Books:

1. K. Ogata, "Modern Control Engineering", Prentice Hall India (2002).
2. Gene F. Franklin, J. D. Powell, A E Naeini, "Feedback Control of Dynamic Systems", Pearson (2008).
3. John Van De Vegte, "Feedback Control Systems", Prentice Hall (1993).
4. Thomas Kailath, "Linear Systems", Prentice Hall (1980).
5. Alok Sinha, "Linear Systems: Optimal and Robust Control", Taylor & Francis (2007).
6. Brian D. O. Anderson and John B. Moore, "Optimal Control: Linear Quadratic Methods", Dover Publications (2007).
7. K. Ogata, "Discrete-Time Control Systems", PHI Learning (2009).
8. H.K. Khalil, "Nonlinear Systems", Prentice Hall (2001).

Course Code: MMTR-215,

Course Name: Machine tool control and condition monitoring,

Credit: 3

Prerequisite NIL

Syllabus:

Overview of automatic control in machine tools

Open loop and closed loop system in machine tools- process model formulation-transfer function control actions-block diagram representation of mechanical pneumatic and electrical systems.

Drive systems and feedback devices in machine tools

Hydraulic and Pneumatic drives, Electrical drives, A.C. Motor, D.C. Motor, Servo motor and Stepper motor. Feedback devices- Synchro, resolver, diffraction gratings, potentiometer, Inductosyn and encoders-application in machine tools.

Adaptive control and PLC: Types- ACC, ACO, Real time parameter estimation, Applications-adaptive control for turning, milling, grinding and EDM. Programmable logic controller- Functions-Applications in machine tools.

Vibration, acoustic emission / sound: Primary & Secondary signals, Online and Off-line monitoring. Fundamentals of Vibration, Sound, Acoustic Emission. Machine Tool Condition Monitoring through Vibration, Sound, Acoustic Emission, Case Studies.

Condition monitoring, through other techniques: Visual & temperature monitoring, Leakage monitoring, Lubricant monitoring, condition monitoring of Lube and Hydraulic systems, Thickness monitoring, Image processing techniques in condition monitoring.

References:

1. Mikell P. Groover, "Automation Production system and Computer Integrated Manufacturing", Prentice Hall of India Pvt. Ltd., 2010

2. Sushil Kumar Srivastava, "Industrial Maintenance Management" S.Chand & Company Ltd., New Delhi, 1998.
3. Manfred Weck, "Hand Book of Machine Tools" – Vol.3, John Wiley & Sons, 1995.

Course Code: MMTR-216,

Course Name: Mechatronics in Metrology

Credit: 3

Prerequisite NIL

Syllabus:

Fundamentals concepts in metrology: Introduction to Metrology and Dimensional Metrology, Angular measurements, Form Measurement, Surface Roughness, Contour Measurements, Roundness, Tool wear measurement.

Instruments for dimensional metrology

Digital Small Tool Instruments, Caliper, Micrometer, Height Gauge, Coordinate Measuring Machines, Types, Design and applications, Opto-electronic Devices for Measurement, Vision Measuring Systems, optical comparators Applications in on-line and in-process monitoring systems, 3D Surface roughness measuring instruments, Applications of LASER in Metrology, Roundness measurement, Laser interferometer in machine tool Inspection, use of laser in on-line Inspection, Laser micrometer, Laser Alignment telescope.

References:

1. Groover. M.P, Automation, and Computer Integrated Manufacturing, Prentice Hall Pvt Ltd 2010
2. Ulrich- Rembold, Armbruster and Ulzmann -" Interface technology for computer controlled manufacturing processes ", Marcel Dekker Pub. New York, 1993
3. N RADHAKRISHNAN, P., Computer Numerical Control (CNC) Machines, New Central Book Agency, 1992.
4. Steave Krar and Arthur Gill, CNC Technology and Programming, McGraw-Hill Publishing Company, 1990.
5. Robert. G. Seippel, -"Optoelectronics for Technology and Engineering", Prentice Hall New Jersey, 1989.

Course Code: MMTR-217,

Course Name: Machine vision,

Credit: 3

Prerequisite NIL

Syllabus:

Introduction: Human vision, Machine vision and Computer vision, Benefits of machine vision. Physics of Light, Interactions of light, Refraction at a spherical surface, Thin Lens Equation.

Image acquisition: Scene constraints, Lighting parameters, Lighting sources, Selection, Lighting Techniques, Types and Selection, Machine Vision Lenses and Optical Filters, Specifications and Selection, Imaging Sensors, CCD and CMOS, Specifications, Interface

Architectures, Analog and Digital Cameras, Digital Camera Interfaces, Camera Computer Interfaces, Specifications and Selection, Geometrical Image formation models, Camera Calibration.

Image processing: Machine Vision Software, Fundamentals of Digital Image, Image Acquisition Modes, Image Processing in Spatial and Frequency Domain, Point Operation, Thresholding, Grayscale Stretching, Neighborhood Operations, Image Smoothing and Sharpening, Edge Detection, Binary Morphology Color image processing.

Image analysis: Feature extraction, Region Features, Shape and Size features, Texture Analysis, Template Matching and Classification, 3D Machine Vision Techniques, Decision Making.

Machine vision applications

Machine vision applications in manufacturing, electronics, printing, pharmaceutical, textile, applications in non-visible spectrum, metrology and gauging, OCR and OCV, vision guided robotics, Field and Service Applications, Agricultural, and Bio medical field, augmented reality, surveillance, bio-metrics.

References

1. Emanuele Trucco, Alessandro Verri, “Introductory Techniques For 3D Computer Vision”, First Edition, 2009
2. Rafael C.Gonzales, Richard.E.Woods, “Digital Image Processing Publishers”, Third Edition, 2007
3. Alexander Hornberg, “Handbook of Machine Vision”, First Edition, 2006
4. Eugene Hecht, A.R. Ganesan “Optics”, Fourth Edition, 2001

Course Code: MMTR-218,

Course Name: Design of Mechanisms and machines

Credit: 3

Syllabus:

Prerequisite- NIL

Mobility analysis, Degree of Freedom (DOF), Mixed Mobility, Total, Partial and Fractional DOF, Closed and Open Chain Systems, Application of D-H representation for Kinematic parameter tables for standard robot structures.

Link coordinate diagram and arm matrix of SCARA, Alpha-II, PUMA articulated robot, standard robot, polar frame, structure robot, Enter transform solution, Arm matrix of standard Robots, Polar frame, structure robots

Structural Analysis and Synthesis of mechanisms, Alternative design solutions; Coding, evaluation and selection of optimum mechanism. Type synthesis, number synthesis and design of mechanisms. Indexes of merit; Graphical, Algebraic and Optimization techniques, Matrix methods of design and analysis; Design of function, Path and Motion Generators; Structural and Mechanical error; Design and Analysis using software like ADAMS. Manipulators-Classifications, actuation and transmission systems; Coordinate Transformation – DH notations, Inverse and Forward kinematics, Manipulator dynamics from Lagrangian and Newtonian point of view. Forces in Manipulator, manipulate Dynamics, selecting of robots for Robot Application Reliability of Robotic & Automation systems and their evaluation.

References:

1. Andeen, G.B., "Robot Design Hand Book", SRI International, McGraw Hill,
2. Craig, J.J., "Introduction to Robotics", Mechanics and Control, Addison Wesley
3. Spong, M., and Vidyasagar, M. "Robot Dynamics and Control", John Wiley, NY, 1989.
4. Venkataraman. S.T., and Liberall. T., "Dextrous Robot Hands", S
5. Appukuttan, "Robotics", I.K. International Publishing house

Course Code: MMTR-219,

Course Name: Artificial Intelligence & Expert System in Automation,

Credit: 3

Prerequisite NIL

Syllabus:

Expert system. Architecture, knowledge base, inference engine, expert system shell, applications.

Fuzzy Logic: Fuzzy sets, membership functions, operation on fuzzy sets; fuzzy control system, Fuzzification, knowledge base, inference, defuzzification, application.

Neural Network: Neuron structure, classification, artificial neural network, back propagation training and algorithm, neuro-fuzzy controllers, applications.

Genetic algorithms: Concepts, encoding and selection methods, genetic operators (crossover and Mutation), applications.

References:

1. Haykin "Neural Networks – A comprehensive Foundation" (Mc-millan)
2. J.M. Zureda "Introduction to artificial neural networks" (Jaico)
3. A Cichocki & R Unbehauen "Neural Networks for optimization and signal Processing" John Wiley
4. George J. Klein & Tina A Polgar "Fuzzy sets, uncertainty and Information"
5. BaertKosko "Neural network and fuzzy systems"
6. Peterson "Introduction to Artificial Intelligence and expert system (PHI)
7. Michell "Introduction to Genetic Algorithm" (PHI)
8. Vidyasagar M "Theory of learning and generalization" Springer
9. S. Rajasekaran, G.A. VijayalakshmiPai "Neural Networks, Fuzzy Logic and Genetic Algorithm", PHI.
10. T.J. Ross: "Fuzzy Logic with Engineering Applications" Second Edition John Wiley India.

Course Code: MMTR-220,

Course Name: Modelling and Simulation of Mechatronic Systems,

Credit: 3

Syllabus:

Prerequisite NIL

Physical Modelling: Mechanical and electrical systems, physical laws, continuity equations, compatibility equations, system engineering concept, system modelling with structured analysis, modelling paradigms for mechatronic system, block diagrams, mathematical models, systems of differential-algebraic equations, response analysis of electrical systems, thermal systems, fluid systems, mechanical rotational system, electrical-mechanical coupling.

Simulation Techniques: Solution of model equations and their interpretation, zeroth, first and second order system, solution of 2nd order electro-mechanical equation by finite element method, transfer function and frequency response, non-parametric methods, transient,

correlation, frequency, Fourier and spectra analysis, design of identification experiments, choice of model structure, scaling, numeric methods, validation, methods of lumped element simulation, modelling of sensors and actuators, hardware in the loop simulation (HIL), rapid controller prototyping, coupling of simulation tools, simulation of systems in software (MATLAB, LabVIEW) environment.

References:

1. L. Ljung, T. Glad, “Modeling of Dynamical Systems”, Prentice Hall Inc. (1994).
2. D.C. Karnopp, D.L. Margolis and R.C. Rosenberg, “System Dynamics: A Unified Approach”, 2nd Edition, Wiley-Interscience (1990).
3. G. Gordon, “System Simulation”, 2nd Edition, PHI Learning (2009).
4. V. Giurgiutiu and S. E. Lyshevski, “Micromechatronics, Modeling, Analysis, and Design with MATLAB”, 2nd Edition, CRC Press (2009).

Course Code: MMTR-221

Course Name: Advanced microcontroller and embedded systems

Credit: 3

Syllabus:

Prerequisite: Microcontrollers and embedded systems

Introduction to embedded systems: Definitions – Brief overview of micro-controllers - DSPs,- Typical classifications –Memory Devices and application scenarios of embedded systems.

Arm 7 core: Introduction about ARM 7 Processor- Internal Architecture – Modes of Operations – Register set –Instruction Sets – ARM Thumb -.Thumb State Registers – Pipelining – Applications.

Arm 9 core: Introduction about ARM 9 Processor--DSP Processor—Sharc Processor -- Internal Architecture – Modes of Operations – Register set – Pipelining – AMBA - Applications.

Real time models, language and operating systems: Models and languages – State Machine and state tables in embedded design – High level language descriptions - Java based embedded system design – Petrinet models-Real time languages – The real time Kernel - OS tasks - Task Scheduling - kernel services – Real time languages and their features.

Case studies in real time embedded systems: Specific examples of time-critical and safety-critical embedded systems - applications in automation -automotives – aerospace - medical and manufacturing.

References

1. Wayne Wolf, Computers as Components – Principles of Embedded Computing System Design, Morgan Kaufmann Publishers 2009.
2. Ball S.R., Embedded microprocessor Systems – Real World Design, Prentice Hall, 2006
3. C.M. Krishna, Kang G. Shin, Real Time systems, McGraw Hill 2009
4. Frank Vahid and Tony Givagis, Embedded System Design
5. Tim Wilmshurst, An Introduction to the design of small – scale Embedded Systems.

Course Code: MMTR-222

Finite element method

Credit: 3

Syllabus:

Prerequisite NIL

Introduction to Finite Difference Method and Finite Element Method, Advantages and disadvantages, Mathematical formulation of FEM, Variational and Weighted residual approaches, Shape functions, Natural co-ordinate system, Element and global stiffness matrix, Boundary conditions, Errors, Convergence and patch test, Higher order elements. Application to plane stress and plane strain problems, Axi-symmetric and 3D bodies, Plate bending problems with isotropic and anisotropic materials, Structural stability, other applications e.g., Heat conduction and fluid flow problems. Idealisation of stiffness of beam elements in beam-slab problems, Applications of the method to materially non-linear problems, Organisation of the Finite Element programmes, Data preparation and mesh generation through computer graphics, Numerical techniques, 3D problems, FEM an essential component of CAD, Use of commercial FEM packages, Finite element solution of existing complete designs, Comparison with conventional analysis.

References:

1. Introduction to Finite Elements in Engineering, T. R. Chandrupatla & A. D. Belegundu, 2nd Ed., PHI, 2001.
2. An Introduction to the Finite Element Method, Reddy, J. N., 2005.
3. The Finite Element Method, O. C. Zienkiewicz, 3rd Ed., McGraw-Hill, 1997.
4. The Finite Element Method in Engineering, S. S. Rao, 2nd Ed., Elmsford, Pergamon, 1989.
5. Introduction to the Finite Element Method: A Numerical Method for Engineering Analysis, Desai, C. S. and Abel, J. F., 1972
6. The Finite Element Method O.C. Zienkiewicz and R.L. Taylor McGraw Hill
7. Finite Element Procedure in Engineering Analysis K.J. Bathe McGraw Hill
8. Finite Element Analysis C.S. Krishnamoorthy Tata McGraw Hill
9. Concepts and Application of Finite Element Analysis R.D. Cook, D.S. Malcus and M.E. Plesha John Wiley
10. Finite Element and Approximation O.C. Zenkiewicz & Morgan

Course Code: MMTR-223

Course Name: Mechatronics in CNC

Credit: 3

Prerequisite NIL

Syllabus:

Computer control of machines

Classification of machine tools, types, functions and processes, Need for CNC in Machine Tools, CNC systems - configuration of the CNC system, interfacing, monitoring, diagnostics, machine data - compensations for machine accuracies Open and closed loop control Adaptive control - types, application and benefits - general configuration of adaptive control and function, reasons

for process change - practical problems with adaptive control - example for feedback and adaptive control.

Mechatronics elements in CNC machine tools

Guide ways and types - mechanical transmission elements - elements for rotary motion to linear motion - ball screw and types - roller screw and types - rack and pinion - various torque transmission elements - requirements of feed drives and spindle drive Types of positional control, Control of Spindle speed, Control of slide movement and velocity. Identification and gauging of work piece. Tool locking system - ball lock mechanism and contact pressure monitoring. Automatic tool changing system - types and benefits - tool magazine.

CNC programming

M/C axis identification, Primary and Secondary, Manual Programming, Turning and Milling Programming fundamentals, fixed cycles, subroutines and other Productivity Packages- CAPP and APT in programming.

REFERENCES:

1. Groover. M.P, Automation, and Computer Integrated Manufacturing, Prentice Hall Pvt Ltd 2010
2. Ulrich- Rembold, Armbruster and Ulzmann -" Interface technology for computer controlled manufacturing processes ", Marcel Dekker Pub. New York, 1993
3. N RADHAKRISHNAN, P., Computer Numerical Control (CNC) Machines, New Central Book Agency, 1992.
4. Steave Krar and Arthur Gill, CNC Technology and Programming, McGraw-Hill Publishing Company, 1990.
5. Robert. G. Seippel,-"Optoelectronics for Technology and Engineering", Prentice Hall New Jersey, 1989.

Course Code: MMTR-224

Theory of Machining

Credit: 3

Syllabus:

Prerequisite NIL

Need and Classifications of Machining Processes; Types of Study in Machining Processes- Experimental and Theoretical; conventional machining: Mechanics of Metal Cutting: Chipping action; Cutting parameters; Orthogonal and Oblique cutting; Mechanism and Types of chips; Cutting forces and Stresses; Power and Energy; Heat and Temperature; Tool Materials and Tool Life; Tool Geometry and Mechanics of Turning, Drilling and Milling; abrasive machining: Mechanics of Metal Grinding; Grinding Wheel and Chip Formation; Grinding Forces and Power; Grinding Temperature; Advanced Abrasive Processes.

References:

1. Materials and Processes in Manufacturing, Degarmo, J. T. Black, Prentice Hall Of India Pvt Ltd.
2. Manufacturing Processes for Engineering Materials, Kalpakjian and Schmid, Prentice Hall.

3. Principles of Machine Tools, G. C. Sen and A. Bhattacharya, New Central Book Agency
4. Theory of Metal Cutting, A. Bhattacharya.
5. Manufacturing Science : Ghosh and Mallick, East-West Press Private Limited
6. Machining and Metal Working Handbook, Ronal A Walsh and Denis Cormier McGraw Hill Publication.

Course Code: MMTR-225

Computer Aided Manufacturing

Credit: 3

Syllabus:

Prerequisite NIL

Basic principles of automation; Extending the capabilities of conventional machines through improved devices and manipulators; Basic principles of numerical control; CNC, DNC and Machining Centres; Manual part programming(industrial controller like Fanuc/Siemens); APT programming; System devices; Adaptive control; Economics of numerical control.

Introduction to Robotics: Elements of robot anatomy; Hydraulic, pneumatic and electrical manipulators; End-effectors and their design, applications of industrial robots.

Reference:

1. CAD/CAM Theory and Practice, Ibrahim Zeid, Tata McGraw Hill Publication
2. CAD/CAM Principle and Application, P. N. Rao, Tata McGrawHill Publication.
3. Users guides to Rapid Prototyping, Todd Frimm, Society of Manufacturing Engineers
4. Computer control of Manufacturing system, Yoram koren, McGraw Hill Publication.
5. Machining and Metal Working Handbook, Ronal A Walsh and Denis Cormier McGraw Hill Publication.
6. Machining and CNC Technology, M. Fitzpatrick, McGraw-Hill Publication.
7. Journey from Rapid Prototyping to Rapid Manufacturing, S Chattopadhyaya, Lambert Publication.

Course Code: MMTR-226

Course Name: Flexible Manufacturing System

Credit: 3

Syllabus:

Prerequisite NIL

FMS concept, Components of FMS, FMS Layouts, FMS planning and implementation. Tool Management systems-Tool monitoring, Work holding devices Modular fixturing, flexible fixturing,, flexibility, quantitative analysis of flexibility, application and benefits of FMS, automated material handling system ,AGVs, Guidance methods, AS/RS.

Automated process planning: Group Technology ,Part families, Part classification and coding, Production flow analysis, Machine cell design, Applications and Benefits of Group Technology, Structure of a Process Planning, Process Planning function, CAPP - Methods of CAPP, CAD based Process **Planning, Inventory management:** Materials requirements planning - basics of JIT.

Monitoring and quality control: Types of production monitoring system, process control & strategies, direct digital control - Supervisory computer control - computer aided quality control -

objectives of CAQC, QC and CIM, contact, non-contact inspection methods, CMM and Flexible Inspection systems. Integration of CAQC with CIM.

References:

1. Kant Vajpayee. S., “Principles of Computer Integrated Manufacturing”, Prentice Hall of India, 1999.
2. Radhakrishnan.P, Subramanyan. S, “CAD/CAM/CIM”, New Age International publishers, 2000.
3. Scheer. A.W., “CIM- Towards the factory of the future” Springer - Verlag, 1994.
4. Daniel Hunt.V., “Computer Integrated Manufacturing Hand Book”, Chapman & Hall, 1989.
5. Groover M.P, “Computer Aided Design and Manufacturing”, Prentice Hall of India, 1987.
6. Yorem Koren, “Computer Control of Manufacturing System”, McGraw Hill, 1986.
7. Ranky Paul. G., “Computer Integrated Manufacturing”, Prentice Hall International, 1986

Course Code: MMTR-227

Rapid Prototyping

Credit: 3

Syllabus:

Prerequisite NIL

Introduction to RP processes, applications, benefits, limitations, comparison between RP processes and CNC Machining.

Different Rapid Prototyping Processes – namely Stereolithography (STL), Laser Sintering; Fused Deposition Modelling (FDM), Solid Ground Curing (SGC); Laminated Object Manufacturing (LOM) , Ballistic Particle Manufacturing (BPM), Three Dimensional Printing (3DP), numerical Examples. **Materials of RP Technology** - Photo Sensitive Resin, Wax etc.

RP interfacing with Manufacturing processes - Investment Casting, Vacuum Casting, and Laser Additive Manufacturing. Rapid manufacturing processes like - Vacuum Casting Processes via RP Processes, Electroplated Prototypes, RTV Silicon Moulds, Direct Inkjet SL Tooling, Electron Beam Melting

References:

1. Rapid Manufacturing: An Industrial Revolution for the Digital Age. Neil Hopkinson, Richard Hague, Philip Dickens (Editors); Wiley; Jan., 2006; ISBN:10: 0470016132; 13: 978-0470016138.
2. Additive Manufacturing Technologies; Rapid Prototyping to Direct Digital Manufacturing. Ian Gibson, David W. Rosen, Brent Stucker; Springer; January, 2010; ISBN: 978-1-4419-1119-3.
3. Rapid Prototyping: Principles and Applications. Rafiq I. Noorani; Wiley; Oct., 2005; ISBN: 10: 0471730017; 13: 978-0471730019.
4. User's Guide to Rapid Prototyping. Todd Grimm; Society of Manufacturing Engineers; February, 2004; ISBN: 0-87263-697-6.
5. Rapid Prototyping - Laser-based and Other Technologies. Patri K. Venuvinod and Weiyin Ma; Kluwer Academic Publishers; October, 2003; ISBN: 1-4020-7577-4.
6. Rapid Prototyping. Andreas Gebhardt; Hanser Gardner Publications; 1st ed., June 1, 2003; (Originally published in German, 1995); ISBN: 156990281X.

7. Rapid Prototyping: Principles and Applications (2nd Edition). Chua Chee Kai, Leong Kah Fai, Lim Chu-Sing; World Scientific Pub Co; March, 2003; ISBN: 9812381171.
8. Rapid Prototyping: Theory and Practice. Ali Kamrani, Emad Abouel Nasr (Editors); Springer; 1st ed., Jan., 2006; ISBN:10: 0387232907; 13: 978-0387232904.
9. Rapid Manufacturing: The Technologies and Applications of Rapid Prototyping and Rapid Tooling, D.T. Pham, S.S. Dimov

Course Code: MMTR-228

MEMS and NEMS

Credit: 3

Syllabus:

Prerequisite NIL

Introduction to MEMS & materials, fabrication processes, MEMS sensors and actuators, scaling issues in MEMS, electro mechanics, Evolution of Microsensors & MEMS, Microsensors & MEMS applications, Microelectronic technologies for MEMS, Micromachining Technology – Surface and Bulk Micromachining, Micromachined Microsensors, Mechanical, Inertial, Biological, Chemical, Acoustic, Microsystems Technology, Integrated Smart Sensors and MEMS, Interface Electronics for MEMS, MEMS Simulators, modelling and design techniques, packaging and reliability, measurement techniques for MEMS, Conclusions & Future Trends.

Nanoelectromechanical systems (NEMS) – a journey from MEMS to NEMS, MEMS vs. NEMS, MEMS based nanotechnology – fabrication, film formation and micromachining, NEMS physics – manifestation of charge discreteness, quantum electrodynamical (QED) forces, quantum entanglement and teleportation, quantum interference, quantum resonant tunneling and quantum transport, Wave phenomena in periodic and aperiodic media – electronic and photonic band gap crystals and their applications, NEMS architecture, Surface Plasmon effects and NEMS fabrication for nanophotonics and nanoelectronics, Surface Plasmon detection – NSOM/SNOM.

References:

1. Electromechanical Sensors and Actuators, Ilene J. Busch-Vishniac, Springer, 2008
2. Introduction to Microelectronics Fabrication, Vol. V, G. W. Neudeck and R. F. Pierret (eds.), Addison, Wesley, 1988
3. Introduction to Microelectromechanical Microwave Systems, H. J. De Loss Santos, 2nd edition, Norwood, MA: Artech, 2004
4. Microsystems Design, S. D. Senturia, Kluwer – Academic Publishers, Boston MA, 2001.
5. Principles and Applications of Nano-MEMS Physics, H. J. Delos Santos, Springer, 2008.
6. Materials and Process Integration for MEMS Microsystems, Vol. 9, Francis E. H. Tay, Springer, 2002.
7. MEMS: Introduction and Fundamentals by Mohamed Gad-el-Hak –
8. MEMS mechanical sensors by Stephen Beeby
9. Microsensors, MEMS, and smart devices by Julian W. Gardner, V. K. Varadan, Osama O. Awadelkarim
10. MEMS and microsystems: design, manufacture, and nanoscale engineering- Tai-Ran Hsu
11. MEMS by N P Mahalik.

Course Code: MMTR-229,

Course Name: Smart materials for mechatronics applications,

Credit: 3

Syllabus:

Prerequisite NIL

Introduction: Smart materials and their application for sensing and actuation, Mechatronics aspects.

Piezoelectric materials: Piezoelectricity and piezoelectric materials, Constitutive equations of piezoelectric materials, Piezoelectric actuator types, Control of piezoelectric actuators, Applications of piezoelectric actuators for precise positioning and scanning.

Shape memory alloys (SMA): Properties of shape memory alloys, Shape memory effects, Pseudo-elasticity in SMA, Design of shape memory actuator, selection of materials, Smart actuation and control, Applications of SMA in precision equipment for automobiles, trains and medical devices

Electro-active polymers (EAPs): Ionic polymer metal composites (IPMC), Conductive polymers, Carbon nanotubes, Dielectric elastomers, Design & control issues for EAP actuators, Applications of EAP for biomimetic, tactile display and medical devices.

Magnetostrictive materials: Basics of magnetic properties of materials, magnetostriction: constitutive equations, types of magnetostrictive materials, Design & control of magnetostrictive actuators, Applications of magnetostrictive materials for active vibration control.

Summary, conclusion and future outlook: Comparative analysis of different smart materials based actuators, Conclusions, Future research trend and applications trends of smart materials and smart materials based actuator technology.

References:

1. Jose L. Pons, Emerging Actuator Technologies, a Micromechatronics Approach, John Wiley & Sons Ltd, 2005
2. Ralph Smith, Smart Material Systems: Model Development, SIAM, Society for Industrial and Applied Mathematics, 2005
3. F. Carpi, D. De Rossi, R. Kornbluh, R. Pelrine, P. Sommer-Larsen, Dielectric Elastomers as Electromechanical Transducers, Elsevier, Hungry, 2008,
4. Y. B. Cohen, Electroactive Polymer (EAP) Actuators as Artificial Muscles Reality, Potential and Challenges, SPIE press, USA, 2004.

Course Code: MMTR-230,

Course Name: Optimization Techniques & Design of Experiments,

Credit: 3

Prerequisite NIL

Syllabus:

Single variable non-linear unconstrained optimization: One dimensional Optimization methods, Uni-modal function, elimination method, Fibonacci method, golden section method, interpolation methods- quadratic & cubic interpolation methods.

Multi variable non-linear unconstrained optimization: Direct search method, Univariate Method – pattern search methods – Powell’s – Hook – Jeeves, Rosenbrock search methods –

gradient methods, gradient of function, steepest decent method, Fletcher reeves method, Variable metric method.

Geometric programming: Polynomials – arithmetic – geometric inequality – unconstrained G.P – constrained G.P

Dynamic programming: Multistage decision process, principles of optimality, examples, conversion of final problem to an initial value problem, application of dynamic programming, production inventory. Allocation, scheduling replacement.

Linear programming: Formulation – Sensitivity analysis. Change in the constraints, cost coefficients, coefficients of the constraints, addition and deletion of variable, constraints. Simulation: Introduction – Types – Steps – application – inventory – queuing – thermal system.

Integer programming: Introduction – formulation – Gomory cutting plane algorithm – Zero or one algorithm, branch and bound method.

Stochastic programming: Basic concepts of probability theory, random variables –distributions – mean, variance, Correlation, co variance, joint probability distribution – stochastic linear, dynamic programming.

References:

1. Optimization theory & Applications/ S.S Rao/ New Age International
2. Introductory to operation research/Kasan& Kumar/Springar
3. Optimization Techniques theory and practice / M.C Joshi, K.M Moudgalya/ Narosa Publications.
4. Operation Research/H.A. Taha/TMH
5. Optimization in operations research/R.L Rardin.
6. Optimization Techniques/Benugundu & Chandraputla/Person Asia.

Course Code: MMTR-231,

Course Name: Additive Manufacturing & Tooling,

Credit: 3

Prerequisite NIL

Syllabus:

Introduction: Historical developments, Fundamentals of RP Systems and its Classification, Rapid prototyping processchains, 3D modeling and mesh generation, Data conversion and transmission.

RP Systems: Liquid polymer based rapid prototyping systems, Teijin Seikis' solid form and other similar commercial RP systems, Solid input materials based rapid prototyping systems, laminated object manufacturing (LOM) and fused deposition modelling systems etc., Power based rapid prototyping systems, selective Laser sintering, SoligenDiren's shell production casting (DSPC), Fraunhofer's multiphase jet solidification (MJS) and MIT's 3D printing (3DP) etc.

RP Database: Rapid prototyping data formats, STL format, STL file problems, STL file repair, Network based operations, Digital inspection, Data warehousing and learning from process data.

RP Applications: Development of dies for moulding, RP applications in developing prototypes of products, application in medical fields, Development of bone replacements and tissues, etc., RP materials and their biological acceptability.

References:

1. Rapid Prototyping of Digital Systems: A Tutorial Approach, Hamblen James O, Kluwer Aca
2. Rapid Prototyping: Principles and Applications, Kai Chua Chee, World Science
3. Rapid System Prototyping with Fpgas: Accelerating the Design Process, R C Cofer, Newnes
4. Rapid Prototyping of Digital Systems, James O Hamblen, Springer.

Course Code: MMTR-232,

Course Name: Vetronics,

Credit: 3

Prerequisite NIL

Syllabus:**Fundamentals of vehicle engineering**

Engine , Types , Modern Engines ,Advanced GDI, Turbo-charged engines Transmissions, Chassis systems , Need for Avionics in Civil and Military aircraft and Space systems

Automotive engine control, monitoring and diagnostics systems

Components of Electronic Engine Management, Engine control functions, Engine control modes, Fuel delivery systems, MPFI, Ignition Systems, Diagnostics , Compression Ignition Engines , Emission control Management , Hybrid Power Plants , BAS.

Automotive transmission and safety systems

Transmission control , Autonomous cruise control , Braking control, ABS , Traction control, ESP, ASR , Suspension control , Steering control , Stability control, Parking Assist Systems, Safety Systems, SRS, Blind Spot Avoidance , Auto transmission electronic control, Telematics, Automatic Navigation, Future Challenges.

Aircraft mechatronics

Fundamentals - components of an airplane and their functions - motions of a plane , Inertial Navigation , Sensors - Gyroscope- Principles , Gyro equations, Rate Gyros - Rate integration and free Gyro, Vertical and Directional Gyros, Laser Gyroscopes, Accelerometers. Direct reading compass, Types of actuation systems-Linear and non-linear actuation system, modeling of actuation systems, Performance testing equipments for sensors and actuation systems. Measurement and control of Pressure, temperature fuel quantity, rpm, torque, engine vibration and power. Electrical Power requirement for Military and Civil standards. Satellite navigation - GPS -system description - basic principles -position and velocity determination

Marine mechatronic systems

Basics of Marine Engineering – Marine Propulsion Mechatronics elements in ships, submarines, Variable Buoyancy Systems.

References:

1. William B.Ribbens, “Understanding Automotive Electronics – 7th Edition, Butterworth, Heinemann Wobum, 2004.
2. Robert N Brady, Automotive Computers and Digital Instrumentation, Areston Book Prentice Hall, Eagle Wood Cliffs, New Jersey, 2000..
3. R.K. Jurgen, Automotive Electronics Handbook, McGraw Hill 2nd Edition. 2000
4. Collinson R.P.G. ‘Introduction to Avionics’, Chapman and Hall, 2002
5. Pallet, E.H.J. ‘Aircraft Instruments & Integrated systems’, McGraw-Hill, 2002
6. Myron Kyton, Walfred Fried, ‘Avionics Navigation Systems’, John Wiley& Sons, 2000

7. Pallett, E.H.J. 'Aircraft instruments, principles and applications', Pitman publishing Ltd., London, 1995

Course Code: MMTR-233,

Course Name: Computational fluid dynamics,

Credit: 3

Prerequisite NIL

Syllabus:

Concept of Computational Fluid Dynamics: Different techniques of solving fluid dynamics problems, their merits and demerits, governing equations of fluid dynamics and boundary conditions, classification of partial differential equations and their physical behavior, Navier-Stokes equations for Newtonian fluid flow, computational fluid dynamics (CFD) techniques, different steps in CFD techniques, criteria and essentialities of good CFD techniques.

Finite Difference Method (FDM): Application of FDM to model problems, steady and unsteady problems, implicit and explicit approaches, errors and stability analysis, direct and iterative solvers. Finite Volume

Method (FVM): FVM for diffusion, convection-diffusion problem, different discretization schemes, FVM for unsteady problems.

Prediction of Viscous Flows: Pressure Poisson and pressure correction methods for solving Navier-Stokes equation, SIMPLE family FVM for solving Navier-Stokes equation, modelling turbulence.

CFD for Complex Geometry: Structured and unstructured, uniform and non-uniform grids, different techniques of grid generations, curvilinear grid and transformed equations. Lattice Boltzman and **Molecular Dynamics:** Boltzman equation, Lattice Boltzman equation, Lattice Boltzman methods for turbulence and multiphase flows, Molecular interaction, potential and force calculation, introduction to Molecular Dynamics algorithms.

Reference Books:

1. J. D. Anderson, "Computational Fluid Dynamics", McGraw-Hill Inc. (1995).
2. S. V. Patankar, "Numerical Heat Transfer and Fluid Flow", Hemisphere Pub. (1980).
3. K. Muralidhar, and T. Sundarajan, "Computational Fluid Flow and Heat Transfer", Narosa (2003).
4. D. A. Anderson, J. C. Tannehill and R. H. Pletcher, "Computational Fluid Mechanics and Heat Transfer", Hemisphere Pub. (1984).
5. M. Peric and J. H. Ferziger, "Computational Methods for Fluid Dynamics", Springer (2001).
6. H. K. Versteeg and W. Malalaskera, "An Introduction to Computational Fluid Dynamics", Dorling Kindersley (India) Pvt. Ltd. (2008).
7. C. Hirsch, "Numerical Computation of Internal and External Flows", Butterworth-Heinemann, (2007).
8. J. M. Jaile, "Molecular Dynamics Simulation: Elementary Methods", Willey Professional, 1997.
9. A. A. Mohamad, "Lattice Boltzman Method: Fundamentals and Engineering Applications with Computer Codes", Springer (2011).

Course Code: MMTR-234,

Course Name: Intelligent visual surveillance systems,

Credit: 3

Syllabus:

Prerequisite: Digital Signal Processing

Basics of Image Processing: Introduction to Image Processing methods, Image Transforms, Wavelet Transform, JPEG Image Compression, Image Formats, Color Spaces- RGB, CMY, HSI.
Video Compression Standards: H. 261, H. 263, H.264, MPEG-1, MPEG-2, MPEG-4, MPEG-7, and MPEG-21, Video shot boundary detection, motion modeling and segmentation techniques.

Object Detection and Classification- Shape based object classification, motion based object classification, Silhouette-Based Method for Object Classification, Viola Jones object detection framework, Multiclass classifier boosting.

Multi-Object Tracking- Classification of multiple interacting objects from video, Region-based Tracking, Contour-based Tracking, Feature-based Tracking, Model-based Tracking, Hybrid Tracking, Particle filter based object tracking, Mean Shift based tracking, Tracking of multiple interacting objects.

Human Activity Recognition- Template based activity recognition, Sequential recognition approaches using state models (Hidden Markov Models), Human Recognition Using Gait, HMM Framework for Gait Recognition, Description based approaches, Human interactions, group activities, Applications and challenges.

Camera Network Calibration - Types of CCTV (closed circuit television) camera- PTZ (pan-tilt zoom) camera, IR (Infrared) camera, IP (Internet Protocol) camera, wireless security camera, Multiple view geometry, camera network calibration, PTZ camera calibration, camera placement, smart imagers and smart cameras.

References:

1. Murat A. Tekalp, "Digital Video Processing", Prentice Hall, 1995.
2. Y. Ma and G. Qian (Ed.), "Intelligent Video Surveillance: Systems and Technology", CRC Press, 2009.

Course Code: MMTR-235,

Course Name: Bio-Mechatronics,

Credit: 3

Syllabus:

Prerequisite: NIL

Bio Mechanics Cardiovascular biomechanics, Musculoskeletal and orthopedic biomechanics, human ergonomic, Rehabilitation. Bio Sensors and Actuators.

Introduction to Bio mechatronics, Electrodes - Types, - Measurement of blood pressure - Blood Gas analyzers: pH of blood, Smart actuators for biological applications.

Medical Measurements Heart rate - Heart sound -Pulmonary function measurements -spirometer -finger-tip oximeter - ESR, GSR measurements.

Signal Processing Bio-medical signals, Signal acquisition and signal processing-Isolation barriers, Bio-Image processing.

Sensory Assist Devices Hearing aids – Implants, Optical Prosthetics, Visual Neuroprostheses – Sonar based systems, Respiratory aids, tactile devices for visually challenged.

Active and Passive Prosthetic Limbs Introduction to prosthetics, Passive Prosthetics – walking dynamics, Knee and foot prosthesis. Active prosthesis - Control of Prosthetic Arms and Hands, Leg Mechanisms, Ankle–Foot Mechanisms, Prosthesis Suspension
Wearable mechatronics devices Wearable Artificial Kidney, Wireless capsule endoscope, Wearable Exoskeletal rehabilitation system, Wearable hand rehabilitation,

References:

1. Graham M. Brooker, “Introduction to Bio-Mechatronics”, Sci Tech Publishing, 2012.
2. Leslie Cromwell, Fred J. Weibell, Erich A. Pfeiffer, “Bio-Medical Instrumentation and Measurements”, II edition, Pearson Education, 2009.
3. Raymond Tong Kaiyu . “Bio-mechatronics in Medicine and Healthcare” Pan Stanford Publishing, CRC Press, 2011.

Course Code: MMTR-236,

Course Name: Internet of Things and Smart Manufacturing,

Credit: 3

Syllabus:

Prerequisite: NIL

Concept of Internet of Things (IoT), common definitions, IoT applications, and functional view. Internet of Things and Internet Technology Cloud Computing, Semantic Technologies, Networking and Communication Technologies.

Contemporary Manufacturing Paradigms- Architecture of IoT-MS, Integration framework of Real-time manufacturing information, Work logic of IoT-MS.

IoT Enabled Manufacturing System- Architecture of IoT-MS, Integration framework of Real-time manufacturing information, Work logic of IoT-MS.

Cloud based Manufacturing Resource configuration- Concept of cloud manufacturing, Real-time production information perception and capturing, Cloud service selection, Cloud Machine model.

Smart Factory and Smart Manufacturing- Concepts of Industry 4.0 standard, Real-time information based scheduling, capacity planning, material planning, Real-time production monitoring techniques with smart sensors, Configuration of smart shop floor, traceability and call back of defective products

Reference:

1. Yingfeng Zhang, Fei Tao, Optimization of Manufacturing Systems using the Internet of Things, Academic Press- Technology & Engineering, 2016.
2. Jiafu Wan, IztokHumar, Daqiang Zhang, Industrial IoT Technologies and Applications, Springer, 17-Aug-2016.
3. K. Wang, Y. Wang, J.O. Strandhagen, T. Yu, Advanced Manufacturing and Automation V, WIT Press, 2016.
4. OvidiuVermesan and Peter Friess, Internet of Things – From Research and Innovation to Market Deployment, River Publishers, 2014.

SEMESTER-III
CORE COURSES

Course Code: MMTR-301,

Course Name: Research Methodology and Technical Writing,

Credit: 3

Syllabus:

Prerequisite- NIL

Research Methodology:

Research process, types of research, problem identification and hypothesis formulation, Research design, methods of data collection, data analysis, reliability and validity, data presentation, and report preparation.

Introduction to Research Communication:

Grammar and Rhetoric: Sentential and supra sentential structure, Narrative and structuring argument, common error in composition. Reading skills for literature review: Previewing techniques, understanding the gist of an argument, identifying the topic sentence.

Writing skills (Part-I)

Sentence formation, Use of appropriate diction, paragraph and essay writing, coherence and cohesion. Summarizing, paraphrasing, outlining, Non-linear description, Narrative, Instruction and reporting. Descriptive and explanatory, analytical and argumentative writing, enhancing editing skills, punctuation.

Writing skills (Part-II)

Introduction to terminology, concept of research. Preparing research proposal/ Synopsis. Formulating thesis statement. Referencing (all style sheet). Writing Introduction, Footnotes/ Endnotes, Conclusion. Preparing Appendix, Bibliography (all style sheets), and Abstract. Writing acknowledgement. Concept of Keywords, preparing content page/ list of Tables and Figures. Use of classified materials, Plagiarism and copyright materials.

References:

1. Ranjit Kumar, research methodology: a step-by-step guide for beginners, SAGE Publications India Pvt Ltd.
2. C.R. Kothari, Research Methodology: Methods and Techniques, New Age International, 2004.
3. Michael H. Markel. Handbook of technical writing. Bedford/St Martins, 2012
4. Technical Writing and Professional Communication: For Nonnative Speakers of English. Thomas N. Huckin. McGraw Hill 1991.
5. Michael H. Markel. Technical Writing Essentials. St. Martin's Press, 01-May-1988.
6. Sharon J Gerson, Steven M Gerson. Technical writing: Process and product. Fifth edition, Pearson Education.

SEMESTER-IV

Course Code: MMTR-451,

Course Name: THESIS /DISSERTATION,

Credit: 13

Syllabus: The students will continue to work on the problem identified in “Thesis/Dissertation-I” (in Semester III) as per the work plan. The work is continued until all stated objectives and deliverables are met. Student will prepare a comprehensive report containing introduction to the problem, literature review, methodology, results and discussion and conclusion. Research Progress Seminar will be held twice a month for continuous evaluation.